Uriel Benymon

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Research interests: Natural language processing and interpretable machine learning, particularly how language models learn compared with human learners, what drives model predictions, and robust evaluation.

Education

Columbia University Sept 2024 – May 2027
Bachelor of Arts in Information Science (Major), Mathematics (Minor)

Spartanburg Community College Jan 2022 – Jul 2024
Associate of Science, Engineering Track
Graduate with Honors

Publications

- **Uriel Benymon**. *Does Lifestyle Information Improve Hypertension Prediction? A Leakage-Controlled Evaluation on NHANES*. **ML4H 2026, Findings Track** [OpenReview]
- **Uriel Benymon**, Albert Ming Wei, Zhicheng Guo, Xinbei Qiu, Clara Resnick. *System-Prompt Invariance for LLM-as-Judge Evaluation: A Linguistic Robustness Study*. **NeurIPS 2026, LP4FM Workshop** [OpenReview]

Posters & Presentations

- **Uriel Benymon**. *A Comparative and Interpretable Machine Learning Investigation of Hypertension Prediction Using NHANES Data.* Abstract published in an abstract booklet & poster presented, **Columbia University Undergraduate Research Symposium, 2026** [Abstract Booklet]
- **Uriel Benymon**. *Exploring Improvements to B-Tree Index Structures for Database Optimization.* Poster presented, **Columbia GS Research Festival, 2026**. Abstract published in an abstract booklet & poster presented, **Columbia University Undergraduate Research Symposium 2025** [Abstract Booklet]

Research Experience

Bancroft Research Scholar, Columbia University Feb 2026 – Apr 2026
Advisor: Professor Ansaf Salleb-Aouissi

- Developed a hypertension-prediction study using NHANES 2017–March 2020 pre-pandemic data, conducting exploratory data analysis and building a preliminary multi-model ML pipeline
- Independently extended the project into a leakage-controlled, cross-validated study of 8,024 participants with five ML models, hyperparameter tuning, bootstrap inference, and SHAP-based interpretability

RISE Scholar, Columbia University Apr 2025 – Jul 2025
Advisor: Professor Paul Blaer

- Benchmarked PostgreSQL's B-Tree index against its other index types (Hash, BRIN, etc.) using a custom Java/SQL environment across randomized, standardized workloads
- Began testing lightweight B-Tree modifications (node capacity/configuration changes) under concurrent operations to explore performance trade-offs

Independent Research Project

Project Lead, System-Prompt Invariance for LLM-as-Judge Study Jan 2026 – Aug 2026

- Led a five-person research team originating from a Columbia LLM course project; designed system-prompt perturbation experiments and developed a Python evaluation pipeline collecting 324 judgments from three LLM graders on 108 responses from four models.
- Recruited volunteer raters for a blinded human validation (Krippendorff's $\alpha = 0.93$; 98% exact agreement with automated scores) and found that graders' compliance with the required output format ranged from 57% to 100%

Teaching

Teaching Assistant III, Columbia University Jun 2026 – Present

- Introduction to Computer Science and Java (COMS1004)

Course Assistant, Columbia University Jan 2026 – May 2026

- Data Structures in Java (COMS3134)

TRIO SSS Peer Tutor, Spartanburg Community College Oct 2023 – Dec 2025

- Tutored peers in-person and virtually in Calculus I-III, C#, C++, Java, SQL, and other coursework

Service

Reviewer, NeurIPS 2026 Workshop on LP4FM Sept 2026

- Provided reviews for 3 papers in the workshop

Research Peer Advisor, Columbia University Oct 2025

- Advised undergraduate peers on pursuing research opportunities and shared research experience at Columbia GS research and academic events

Awards & Honors

- Columbia GS Honor Society, October 2026
- Bi Ran and Chuck Scholarship (Columbia GS), July 2026
- Columbia GS Service Award, April 2026
- Dean's List, Columbia University & Spartanburg Community College (multiple terms)
- Columbia GS Phi Theta Kappa Scholarship (multiple terms)
- Clemson SPECTRA Scholarship, August 2023
- Phi Theta Kappa Honor Society, April 2023
- National Technical Honor Society, April 2023

Skills

Languages: Python, Java, SQL, C++, C#, C
Tools: pandas, NumPy, scikit-learn, XGBoost, SHAP, SciPy, NLTK, imbalanced-learn, LaTeX